

Solutions to the Quantum Computing Hackathon Télécom Paris

Team 2 Report

February 2025

1 Question 1: To C or not to C

Problem statement: Create the Bell state $\frac{1}{\sqrt{2}}(|0\rangle_0|0\rangle_{19} + |1\rangle_0|1\rangle_{19})$ between qubit 0 and qubit 19 on a quantum machine with a specific connectivity graph. The goal is to minimize the circuit depth after transpilation.

Our solution

First idea

The first exercise is very similar to an exercise seen in the TP the same morning. So naturally our first idea was to copy the idea from the correction, and switch qubit 19 to qubit 2 through SWAP gates, then apply the classic modification to create a Bell+ state, i.e., a Hadamard gate and a CX gate. But the test verifies that at the end qubits 0 and 19 are still in their places. So we had to switch the gate again from 2 to 19. This required 18 SWAP gates + H + CX + 18 SWAP gates. Considering a SWAP gate is 3 CX gates, the depth was very poor.

How we got our solution

By decomposing the SWAP gate as 3 CX gates, we saw that for all of the qubits there were 6 CX gates which simplified to 2 CX gates (since $CX \cdot CX = I$). So we replaced all the SWAP gates by CX gates and used Hadamard on qubit 0. The solution is presented in Figure 1 and the circuit schema in Figure 2.

The optimal path found is: `path = [0, 2, 3, 6, 8, 9, 12, 13, 16, 17, 19]`

Final result: depth = 22

2 Question 2: State Preparation

Problem statement: Prepare the quantum state

$$|\psi\rangle = \frac{1}{\sqrt{8}}(|00000\rangle + |01011\rangle + |01101\rangle + |01110\rangle + |10001\rangle + |10010\rangle + |10100\rangle + |11111\rangle) \quad (1)$$

up to a global phase, using ancilla qubits, single-qubit rotations, and CX gates. The score is $S = 0.99^d + |\langle\psi|\rho|\psi\rangle|$ where d is the circuit depth and ρ is the prepared state.

Our solution

The first thing that we saw is that for each qubit, there was exactly half $|0\rangle$ and half $|1\rangle$. Also, the state is a superposition of 2^3 terms, the qubits are entangled, and the normalization term is $\frac{1}{2\sqrt{2}}$. We immediately thought about using 3 Hadamard gates and CNOTs to link all the qubits

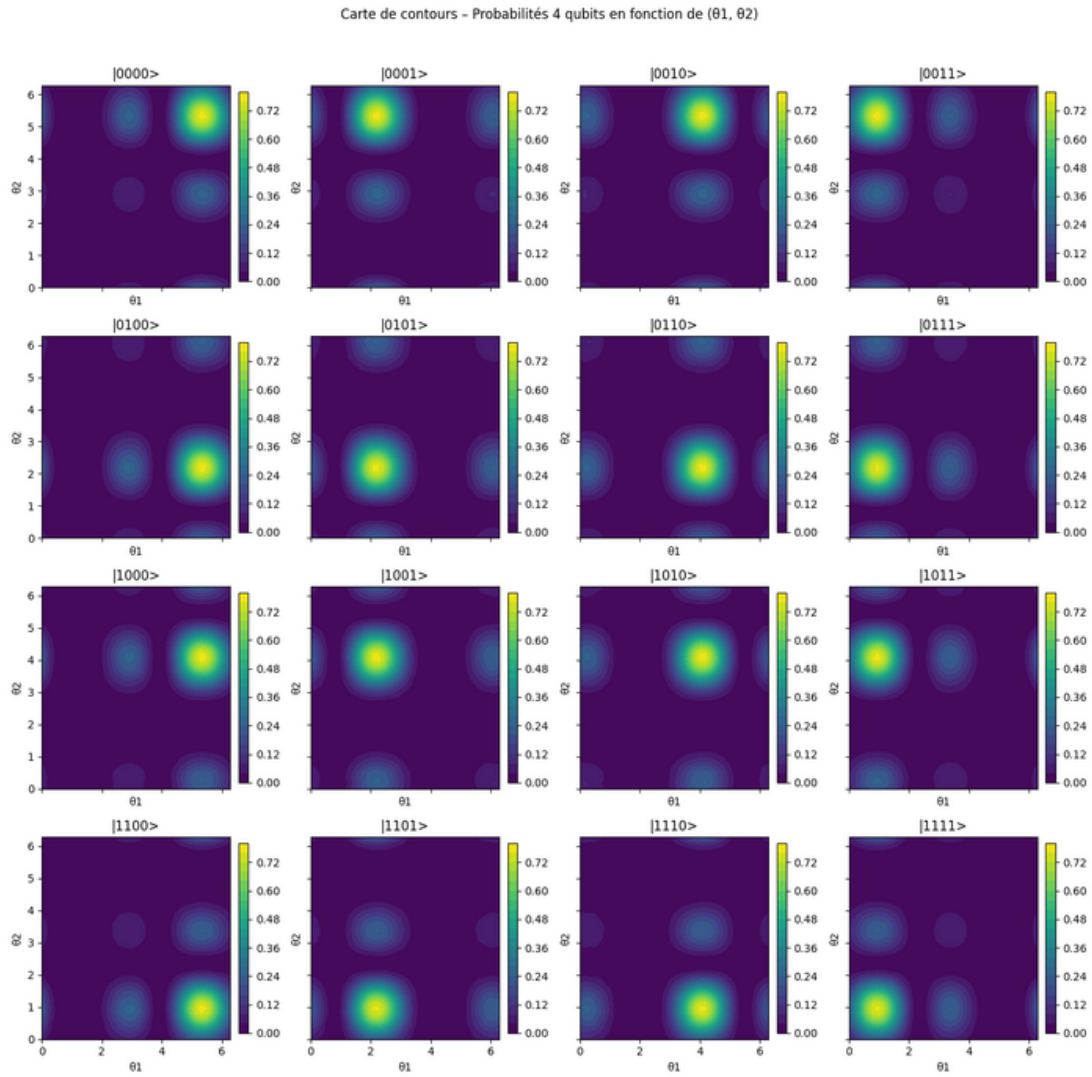


Figure 1: Contour map showing probability optimization for the 4-qubit circuit as a function of parameters (θ_1, θ_2) .

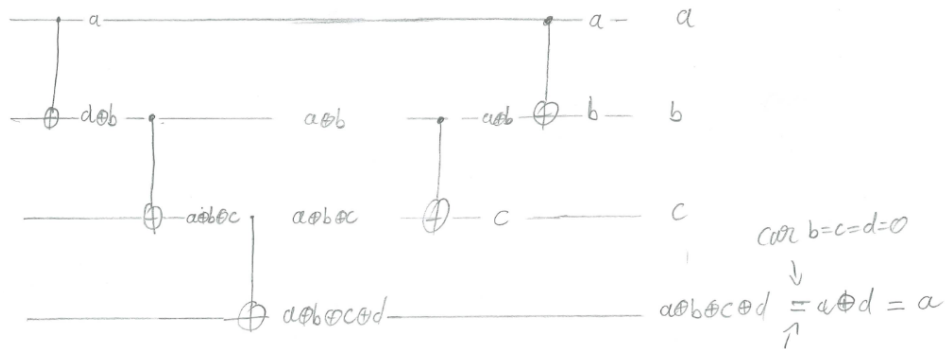


Figure 2: Hand-drawn circuit schema showing the structure of operations with qubits a, b, c, d and XOR operations.

to get such a state, with a term with all $|1\rangle$ and another one with all $|0\rangle$. Thus, we tried to see patterns by analyzing the results that we obtained by placing these gates randomly at first, then adapting the circuit.

Seeing how hard the task really was (we sure underestimated how hard it would be) and how we really struggled to just find the solution without even considering depth, we decided to completely change our strategy.

In our first experimentation with this exercise, we understood that we had to use rotation gates (to create the superposition of states) and CX gates to entangle the states. We also analyzed the scoring function and realized that it could easily give us (with really quick modifications) a loss function.

Given these two facts, we pivoted to a variational algorithm. We kept the idea of layers of rotations with parameters θ, ϕ, λ and CX gates (see the ansatz structure).

We modified the scoring function to give us a loss:

```
def loss(qc, params_list, params_obj, psi):
    ...
    return 1 - np.abs(psi.T.conj() @ dm @ psi)[0, 0]
```

mainly changing the return (no 0.99^d term) to $1 - |\langle\psi|\rho|\psi\rangle|$.

Then, using the code from the TP, we created an optimization function for our parameters. We then tried changing the number of layers to optimize the final score. Increasing the number of layers increased the depth but gave us better precision. Finally, we found that three layers was optimal.

Final result: Score = 1.835

3 Question 3: Random State Preparation

Problem statement: Create a "random" quantum state using only a single parameter $\lambda \in [0, 1]$ on 2 qubits. The state should replicate the first and second moments of the Haar measure for Pauli operators. The score is $S = \frac{1}{15} \sum_{P_1, P_2 \neq II} (|\langle P_1 P_2 \rangle| + (\langle P_1 P_2 \rangle^2 - 1/3)^2)$ where lower is better.

Our solution

The exercise here was to create a "random state", defined here as a state that is the closest to the Haar measure on Pauli operators.

If we try to understand what it means:

- **For the expectation:** We want to create a superposition of states that will be a uniform distribution on the Bloch sphere. (The sum on X gates $\Rightarrow$ uniform for the 0,1 axis, Z for +,- axis, etc.)

So, of course we use rotations around the Bloch sphere (X, Y) with parameter λ . We debugged by printing the results regarding the expectation and did not find good results.

- **For the variance:** We want the states to be dependent, but not too much, so we entangle the states.

The next idea we had was to choose prime numbers (actually very high prime numbers) to break the periodicity (so that a state doesn't make a full turn). We used:

$$\vec{w} = [118927, 118931, 118967, 118973, 119027, 119033, 119039, 119047, 129313, 129341, 129347, 7, 11] \quad (2)$$

We searched manually for what would be the best solution and found good results with 5 layers of X and Y rotations with angles $w_i \pi \lambda$ and CX gates.

Final result: Low score (close to 0)

4 Question 4: Find the Lazy Bit

Problem statement: Given an oracle $U_f : |x\rangle|y\rangle \rightarrow |x\rangle|y \oplus f(x)\rangle$ where $f(x) = 1$ if and only if all three "working" bits among the four input bits are set to 1. One bit is "lazy" and doesn't influence f 's output. The goal is to identify the lazy bit's index with the highest probability using the oracle as few times as possible.

Our solution

Version with classical approach that doesn't leverage superposition

The objective was to identify the "lazy bit" among 4 qubits using an oracle that marks states where all "working bits" are set to 1. The final solution consists of a classical-style quantum approach.

The main problem we encountered is the presence of the $|1111\rangle$ state in the search space. For a given lazy bit index i , the oracle marks two states: the target state x_i where all bits are 1 except the lazy bit at 0, and the $|1111\rangle$ state where all bits are 1. Grover's algorithm amplifies both states equally, which means we get $|1111\rangle$ about 40-50% of the time, a state that doesn't allow us to identify the lazy bit since no bit is 0. We attempted to disfavor $|1111\rangle$ with a few techniques (mainly found on the internet) but none seemed to work.

This problem seemed like a "search" problem, so Grover's algorithm came to mind. We thought we could use phase kickback and amplify all qubits except the lazy one, which suggested a possible $O(\sqrt{n})$ solution. The problem here is that when we prepare a superposition of all the states, thanks to Grover's algorithm we managed to find elements in the set $E = \{\text{qubits with at least one 1}\}$.

So we switched to a classical-style approach: start with all 1s, then for each bit, flip it to 0, apply the oracle, check if $f(x) = 0$, and either return the bit or reset and continue. This is an $O(n)$ method. It worked perfectly with 8 oracle calls, and Omar improved it to just 4.

Here's the idea in the quantum version:

1. Start with $|0\rangle|1111\rangle$ where $|0\rangle$ is the ancilla
2. Flip one bit (say to get $|?0??\rangle$), apply the oracle to see if it satisfies $U_f(x) = 1$
3. The ancilla records the result: $|1\rangle$ if it's the lazy bit, $|0\rangle$ otherwise
4. Flip the bit back
5. If the ancilla is $|1\rangle$, leave that bit at 0. If the ancilla is $|0\rangle$, restore it to 1
6. Repeat for all bits

At the end, the lazy bit is the one left at 0. The circuit is shown in Figure 3.

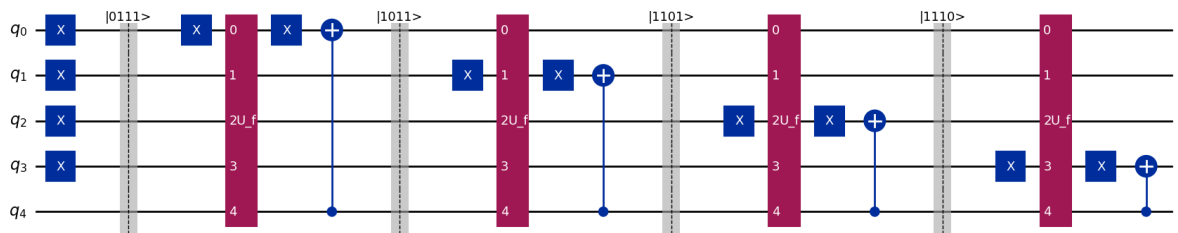


Figure 3: Quantum circuit for the lazy bit identification showing the sequential testing of all four qubits with the oracle U_f .

Final result: Probability of success = 1.000, Number of oracle uses = 4

5 Question 5: Order, Order

Problem statement: Determine whether two single-qubit gates U_1 and U_2 commute ($U_1U_2 = U_2U_1$) or anticommute ($U_1U_2 = -U_2U_1$) using a dirty qubit initialized in a random state $|\psi\rangle$ and clean ancilla qubits initialized to $|0\rangle$.

Our solution

The first idea was to create a circuit without the dirty qubit and to swap a qubit for which we knew that the measurement would give the expected result with it. We thought immediately of the Deutsch-Jozsa algorithm, particularly the interference terms which could give some hints about constant/balanced functions.

The product $U_1U_2U_2^\dagger U_1^\dagger$ is either the identity (constant) or its negative (balanced). Trying to implement it, we saw that it would be difficult because we didn't have a U_f gate. We wanted to do a similar idea, which means using the minus sign created by anticommutation to have a probability equal to 0.

Therefore, we decided to use U_1 , U_2 , and their inverses to create $\pm I$. To have simultaneously $|0\rangle$ and $\pm|0\rangle$, we used an H gate on qubit 1 to have $|+\rangle$ and control differently qubit 0 (dirty qubit) like we do to create a Bell state.

Thus, what we have is:

$$t_0 : |\psi\rangle|0\rangle \quad (3)$$

$$t_1 : \frac{1}{\sqrt{2}}(|\psi\rangle|0\rangle + |\psi\rangle|1\rangle) \quad (4)$$

$$t_2 : \frac{1}{\sqrt{2}}(|\psi\rangle|0\rangle \pm |\psi\rangle|1\rangle) \quad (\text{after applying } C-U_2^\dagger, C-U_1, C-U_2, C-U_1^\dagger) \quad (5)$$

Writing that on paper made us realize that qubit 1 is in the $|+\rangle$ or $|-\rangle$ state, depending on whether the product commutes or not. We simply had to apply an H gate again to get the result that we want after the measurement.

We then swapped qubit 1 with the dirty qubit to measure the right qubit in the scoring cell. In fact, the value of the dirty qubit doesn't matter initially, and we can use it directly to play the role of qubit 1, saving an ancilla qubit.

Figure 4 shows the probability distributions of the states as a function of θ , illustrating the clear separation between commuting and anticommuting cases.

We obtain a probability of success of 100% with 4 gate uses and only one qubit more than the dirty one (actually 0 additional qubits since we use the dirty qubit itself).

Final result: Probability of success = 100%, Number of uses = 4, Number of ancilla qubits = 0

6 Question 6: Ansatz Design

Problem statement: Create a 4-qubit parametrized circuit that can prepare as many computational basis states as possible while minimizing the number of parameters. The score is $S = N_{\text{states}}/N_{\text{parameters}}$ where N_{states} is the number of basis states the circuit can generate.

Our solution

Naive answer (score = 4 = 16/4)

The simplest solution consists of using 4 parameters $\{p_1, p_2, p_3, p_4\}$ with $R_x(p_i)$ rotations on each qubit i . Each parameter can take the value 0 or π to generate $|0\rangle$ or $|1\rangle$ respectively.

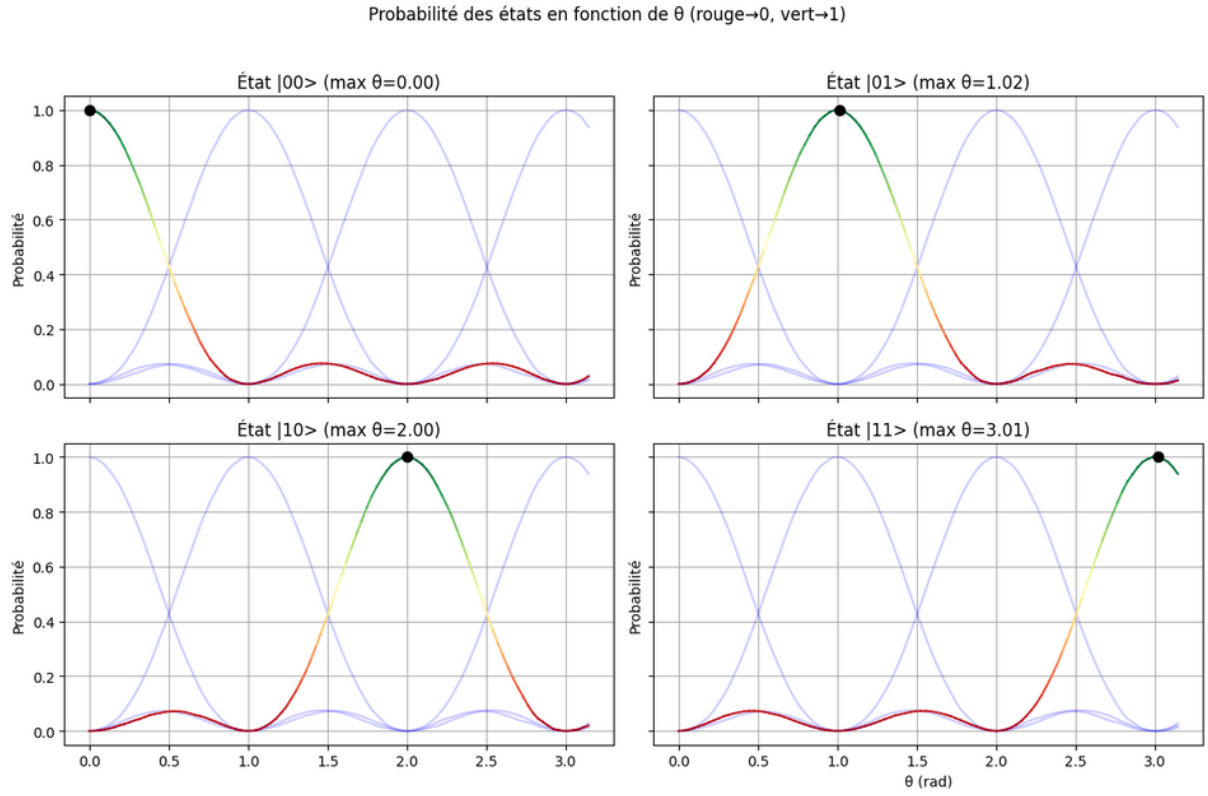


Figure 4: Probability distributions of quantum states as a function of θ (in radians). The four panels show different computational basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, with maximum probabilities indicated.

This allows us to generate all $2^4 = 16$ computational basis states with 4 parameters:

$$\text{answer} = \{f(i) \mapsto \{p_j : ((i \gg j) \& 1) \times \pi\}_{j=1}^4\}_{i=0}^{15} \quad (6)$$

where $f(i)$ formats integer i as a 4-bit binary string.

Final result: Score = 4.000

7 Question 7: A Metrology Question

Problem statement: Estimate the magnetic field strength B that causes the evolution $U(B) = \prod_{i=0}^{14} R_z(B/100)_i$ on 15 qubits after 1 millisecond interaction. The circuit can use $U(B)$ at most once, and the estimate $\hat{B}$ is made from 100 measurements. The score is the average error $\mathbb{E}_B[|\hat{B} - B|]$ with $B \in [0, \pi]$.

Our solution

The first difficulty was to get B out of the phase because it wouldn't show in the probability otherwise. Having a complex amplitude with $|0\rangle$ and its conjugate with $|1\rangle$, our idea was to sum them to create a cos that we could observe.

Thus, we added a Hadamard gate before the $U(B)$ gate to create a superposition and have both complex terms. To sum these terms, we wanted to change only the $|1\rangle$ to a $|0\rangle$ but this is impossible because it is not reversible and the result could not be unitary.

The only way to see the cos is by it being a probability when we measure. Therefore, we used another Hadamard gate after $U(B)$ to sum both exponentials and have a cos in front of the $|0\rangle$ and a sin in front of the $|1\rangle$. Thanks to the measurements, we should be able to know with which probability we obtain $|0\rangle$, this probability being $\cos^2(B/100)$.

We implemented this idea on the first qubit only at first to see what it would give, but the results that we obtained were not satisfying since we had the impression that we always had $P(0) = 1$. We decided to increase our precision by using AI tools, which told us that the issue was that with $B \in [0, \pi]$, we have $B/100 \in [0, \pi/100]$ which is really small, and thus $\cos(B/100) \approx 1$.

The AI assistant gave us the idea to create a GHZ circuit such that we get:

$$P(|0\dots 0\rangle) = \cos^2(15B/200) = \cos^2(3B/20) \quad (7)$$

This increases the sensitivity by a factor of 15.

The GHZ preparation:

1. Apply H on qubit 0
2. Apply CX gates sequentially: $CX_{i,i+1}$ for $i = 0$ to 13

After $U(B)$ application and inverse GHZ operations, we measure all qubits. From the fraction p of $|0\dots 0\rangle$ outcomes, we estimate:

$$\hat{B} = \frac{40}{3} \arccos(\sqrt{p}) \quad (8)$$

Then, we tried to optimize the results with AI by having the best post-treatment to extract B from measurements.

Best result obtained: Score ≈ 0.471

8 Question 8: Which State Did I Send?

Problem statement: One of three qubit states is sent with equal probability $1/3$:

$$|\psi_0\rangle = |0\rangle \quad (9)$$

$$|\psi_1\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (10)$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle - e^{i5\pi/4}|1\rangle) \quad (11)$$

Design a three-qubit unitary where the unknown state enters on qubit 0 and qubits 1, 2 are ancillas. The measurement outcome b_1b_2 indicates: $00 \rightarrow$ guess state 0, $01 \rightarrow$ guess state 1, $10 \rightarrow$ guess state 2, $11 \rightarrow$ "don't know". The score is the probability that the guess is correct.

Our solution

We approached this problem by first identifying it as a quantum state discrimination task of non-orthogonal states, where perfect identification is fundamentally impossible. As an initial step, we did a brief theoretical review to estimate the expected order of magnitude of the optimal success probability, which we found to lie in the range of approximately 0.6–0.7. This confirmed that the objective was to maximize the guessing probability rather than attempt exact reconstruction.

We then analyzed how the three states differ and observed that their distinguishing information is primarily encoded in small phase and amplitude variations that are lost under direct measurement. To make this information accessible, we applied an initial single-qubit $R_y(-\pi/8)$ rotation. This rotation recenters the cluster so that:

- State 1 lies mostly on the "low-excitation" side
- State 2 is near the decision boundary
- State 3 crosses into the opposite lobe

We next used entanglement with an ancillary qubit using a CNOT gate. After the R_y :

- State 1 has low $|1\rangle$ population on qubit 0 $\rightarrow$ qubit 1 stays mostly 0
- States 2 and 3 have higher $|1\rangle$ weight $\rightarrow$ qubit 1 flips more often

So qubit 1 performs a binary test: "Is the state closer to state 1 or not?"

This already explains why: state 1 $\rightarrow 0^*$ and states 2 & 3 $\rightarrow 1^*$. If we removed this, state 1 would collapse into the same bucket as state 2.

To further distinguish the remaining two states, we applied a controlled $R_y(\pi)$ gate, which is effectively a conditional NOT with phase sensitivity. What happens:

- State 2 has partial $|1\rangle$ on qubit 0 $\rightarrow$ qubit 2 flips with moderate probability
- State 3 has larger $|1\rangle$ weight $\rightarrow$ qubit 2 flips almost deterministically
- State 1 barely activates it at all

So qubit 2 answers: "If it's not state 1, which side of the cluster is it on?"

This is why: state 2 $\rightarrow 01$ and state 3 $\rightarrow 10$. They differ mostly on qubit 2, not qubit 1.

Finally, a second CNOT gate was used to concentrate probability mass onto distinct computational-basis outcomes, making a clear correspondence between each input state and a dominant measurement result that approaches the expected theoretical performance.

Final result: High probability of success (≈ 0.65 – 0.70)